WORKED SOLUTION

STEP II 1987, Question 15

Probability and Statistics • Geometric Probability

STEP II 1987, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q15

The key idea

Forget positions and think about *times*. Each vehicle occupies the crossing for an interval of time, and a collision happens exactly when the two intervals overlap. Writing that overlap as two inequalities turns the collision region in the (u_1, u_2)-plane into a wedge between two lines through the origin. With u_1, u_2 uniform on $(0, V)$, the point (u_1, u_2) is uniform on a square, so the probability is just the fraction of the square lying inside the wedge, and the three cases correspond to where the two lines leave the square.

Solution

The collision condition

We model the train and lorry as moving at constant speeds $u_1, u_2 > 0$ along straight roads that cross at a single point (the width of the crossing is neglected, or absorbed into the lengths). A collision occurs if and only if some part of each vehicle is at the crossing at the same instant.

The front of the train reaches the crossing at time d_1/u_1 and its back passes at time $(d_1 + l_1)/u_1$, so the train occupies the crossing for

$$\frac{d_1}{u_1} < t < \frac{d_1 + l_1}{u_1}.$$

Likewise the lorry occupies it for $d_2/u_2 < t < (d_2 + l_2)/u_2$. Two open intervals (a_1, b_1) and (a_2, b_2) overlap exactly when each starts before the other ends: $a_1 < b_2$ and $a_2 < b_1$. So a collision occurs if and only if

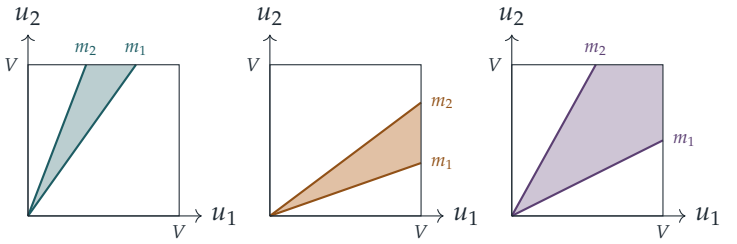
$$\frac{d_1}{u_1} < \frac{d_2 + l_2}{u_2} \quad \text{and} \quad \frac{d_2}{u_2} < \frac{d_1 + l_1}{u_1}.$$

Multiplying through by the positive quantity $u_1 u_2$ and rearranging, this is

$$\boxed{\frac{d_2}{d_1 + l_1}\, u_1 \;<\; u_2 \;<\; \frac{d_2 + l_2}{d_1}\, u_1.}$$

(If an inequality is an equality the vehicles only touch, front to back; this has probability zero, so it doesn't matter whether we count it.)

Write $m_1 = \dfrac{d_2}{d_1 + l_1}$ and $m_2 = \dfrac{d_2 + l_2}{d_1}$. Since $d_2 < d_2 + l_2$ and $d_1 + l_1 > d_1$, we have $m_1 < m_2$, so the collision region is the wedge in the positive quadrant between the lines $y = m_1 x$ and $y = m_2 x$. The pictures below show it, shaded, inside the square $0 < x, y < V$ that we need next.



(A) $d_1 + l_1 \le d_2$ (B) $d_2 + l_2 \le d_1$ (C) otherwise

The probability when the back of the train is nearer than the front of the lorry

Since u_1 and u_2 are independent and each uniform on $(0, V)$, the pair (u_1, u_2) has constant joint density $1/V^2$ on the square $(0, V)^2$. So the probability of a collision is

$$\frac{\text{area of the wedge inside the square}}{V^2}.$$

? **Ask yourself** Which sides of the square do the two lines leave through?

The line $y = mx$ leaves the square through the top side if $m \ge 1$ (at $(V/m, V)$) and through the right-hand side if $m \le 1$ (at (V, mV)). In this case $d_1 + l_1 < d_2$, so $m_1 = d_2/(d_1 + l_1) > 1$, and therefore $m_2 > m_1 > 1$ as well: both lines leave through the top, as in picture (A).

The wedge inside the square is then the triangle with vertices $(0,0)$, $(V/m_2, V)$ and $(V/m_1, V)$. Its base along the top side has length $V/m_1 - V/m_2$ and its height is V, so

$$\begin{aligned}\mathrm{P}(\text{collision}) &= \frac{1}{V^2} \cdot \frac{1}{2} V \left(\frac{V}{m_1} - \frac{V}{m_2} \right) = \frac{1}{2} \left(\frac{d_1 + l_1}{d_2} - \frac{d_1}{d_2 + l_2} \right) \\ &= \frac{(d_1 + l_1)(d_2 + l_2) - d_1 d_2}{2 d_2 (d_2 + l_2)} = \frac{l_1 l_2 + l_2 d_1 + l_1 d_2}{2 d_2 (l_2 + d_2)},\end{aligned}$$

since the $d_1 d_2$ terms cancel. This is the required result.

The other two cases

Back of the lorry nearer than the front of the train, $d_2 + l_2 < d_1$. Now $m_2 = (d_2 + l_2)/d_1 < 1$, so $m_1 < m_2 < 1$ and both lines leave through the right-hand side, as

in picture (B). The wedge is the triangle with vertices $(0,0)$, (V, m_1V), (V, m_2V): base $(m_2 - m_1)V$ on the side $x = V$, height V. So

$$\mathrm{P}(\text{collision}) = \frac{1}{2}(m_2 - m_1) = \frac{1}{2}\left(\frac{d_2 + l_2}{d_1} - \frac{d_2}{d_1 + l_1}\right) = \frac{l_1 l_2 + l_1 d_2 + l_2 d_1}{2d_1(l_1 + d_1)}.$$

This is the first answer with the roles of train and lorry swapped ($l_1 \leftrightarrow l_2$, $d_1 \leftrightarrow d_2$), as it must be, since swapping the vehicles amounts to reflecting the picture in $y = x$.

Neither vehicle entirely ahead of the other, $d_2 \le d_1 + l_1$ and $d_1 \le d_2 + l_2$. Now $m_1 \le 1 \le m_2$, so the line $y = m_1 x$ leaves through the right-hand side and $y = m_2 x$ through the top, as in picture (C). It is easiest to subtract the two non-collision triangles from the square: the one below $y = m_1 x$ has area $\frac{1}{2}V \cdot m_1 V$, and the one to the left of $y = m_2 x$ (between it and the u_2-axis) has area $\frac{1}{2}V \cdot V/m_2$. Hence

$$\mathrm{P}(\text{collision}) = 1 - \frac{m_1}{2} - \frac{1}{2m_2} = 1 - \frac{d_2}{2(d_1 + l_1)} - \frac{d_1}{2(d_2 + l_2)}.$$

The answers agree where the cases meet: at $d_2 = d_1 + l_1$ (so $m_1 = 1$) the first and third formulae both give $\frac{1}{2} - \frac{d_1}{2(d_2 + l_2)}$, and similarly at $d_1 = d_2 + l_2$. Notice that none of the probabilities depend on V: scaling both speeds by the same factor doesn't change whether the vehicles collide.

Checks

- With $l_1 = 1$, $l_2 = 2$, $d_1 = 1$, $d_2 = 5$ (case A), the formula gives $\frac{9}{70} \approx 0.1286$; a simulation of 4×10^5 random pairs of speeds gave 0.1283. Simulations in cases B and C (for example $l_1 = l_2 = 1$, $d_1 = 1$, $d_2 = 1.5$, giving 0.425) agreed to within 0.002.
- Letting $l_1, l_2 \to 0$ in case A gives probability 0: two points meet at the crossing only if $u_2/u_1 = d_2/d_1$ exactly.
- In case C with $d_1 = d_2$ and very long vehicles, $m_1 \to 0$ and $m_2 \to \infty$, so the probability tends to 1, which is sensible.

Common slips

- Writing only one inequality, such as "the train arrives before the lorry has cleared the crossing"; overlap of two intervals needs *both* conditions.
- Using the formula for a triangle in case C, where the region is a quadrilateral; check where each line leaves the square first.

Beyond STEP

The method here, turning a probability for independent uniform variables into an area, is geometric probability, and it is the simplest instance of computing $\mathrm{P}((X, Y) \in A)$ as a double integral of a joint density. The same idea underlies classic problems such as Buffon's needle and the "meeting problem" of two people arriving at random times. Grimmett and Welsh, *Probability: An Introduction*, develops joint densities and their use in this way.

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